Chapter 2

薄膜沉积

Thin film deposition-1

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Outline

Introduction

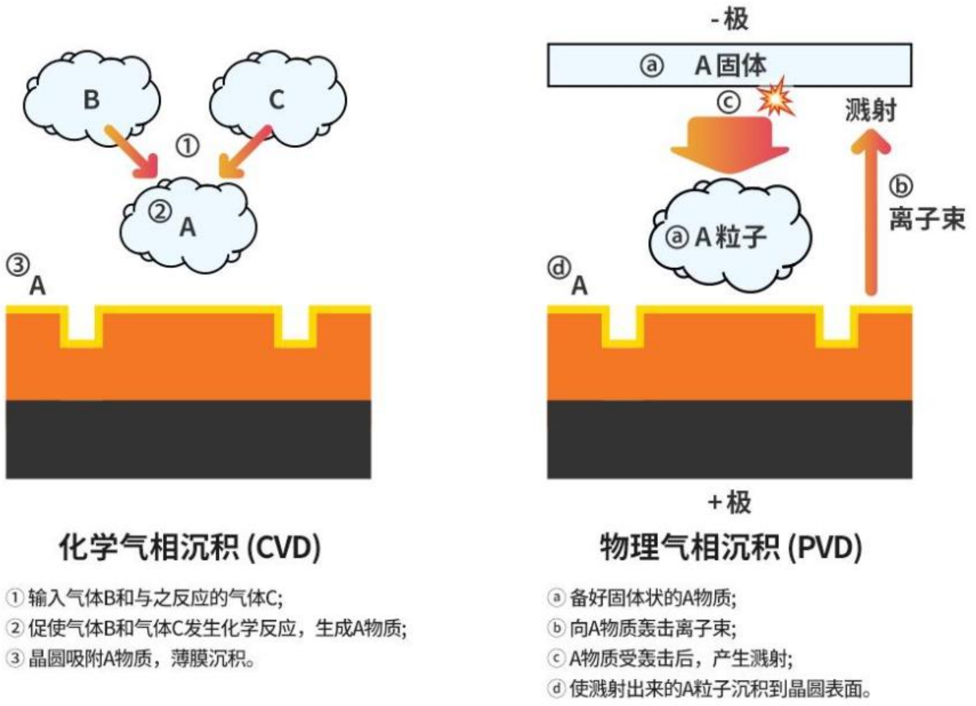
- THIN FILM DEPOSITION
  - CHEMICAL METHODS
    - Sol-gel
      - Dipping
      - Spraying
      - Spin Coating
    - Plating
      - Electroplating
      - Ekectroless
    - Chemical Vapor Deposition
      - MOCVD
      - PECVD
      - LPCVD
      - Thermal
      - ALD
  - PHYSICAL METHODS
    - Evaporation
      - Ion Plating
      - Laser ablation
      - MBE
      - Electron Beam
      - Thermal
    - Sputtering
      - RF
      - DC
      - Magnetron
      - IBS

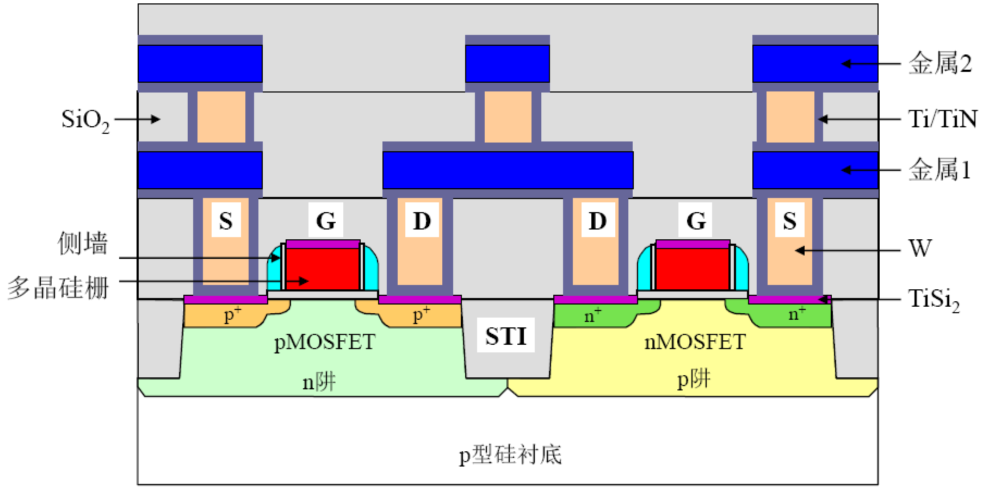
Introduction

- **Materials**
  - metals
  - dielectrics
  - polymers
- **Desired composition**
- **Low contamination**
- **Uniformity**
  - thickness/morphology across sample
  - thickness/morphology run to run
- **Directionality**
  - directional: good for lift-off, trench filling
  - non-directional: good for step coverage

- **Film properties**
  - Stress
  - Adhesion
  - Stoichiometry
  - Density
  - Grain size, boundary properties, orientation
  - Electrical (e.g., breakdown voltage)
  - Optical
  - Magnetic

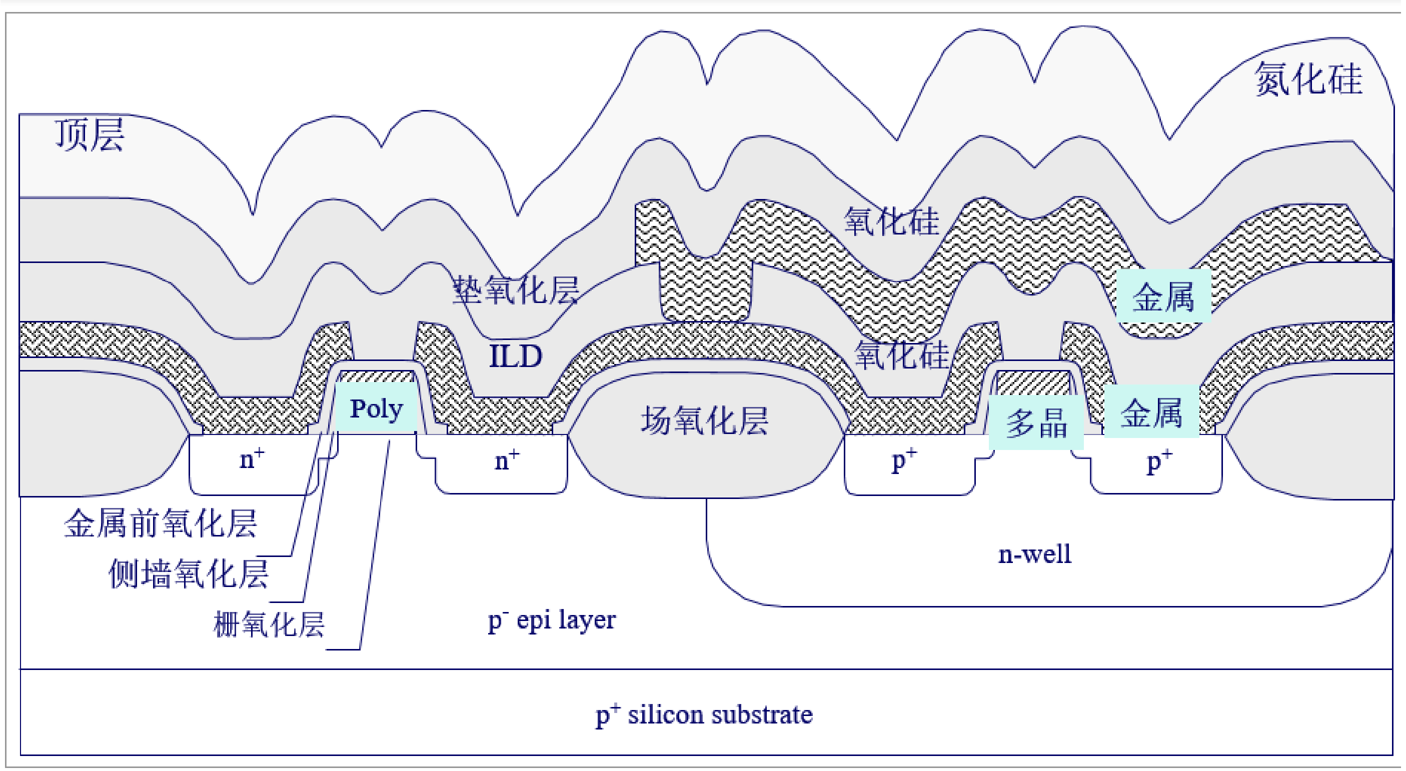
- **Main Deposition Methods**
  - Thermal oxide
  - PVD: sputtering, evaporation
  - CVD
    - Principles
    - Equipment
    - deposited films





多层薄膜沉积

- 表面不平坦，这将成为VLSI多层金属高密度芯片制造的限制因素. 必须考虑台阶覆盖性
- 随着特征尺寸越来越小，层数多，需要更多的金属来做连接
- 各金属之间的绝缘非常重要
- 淀积可靠的薄膜材料至关重要



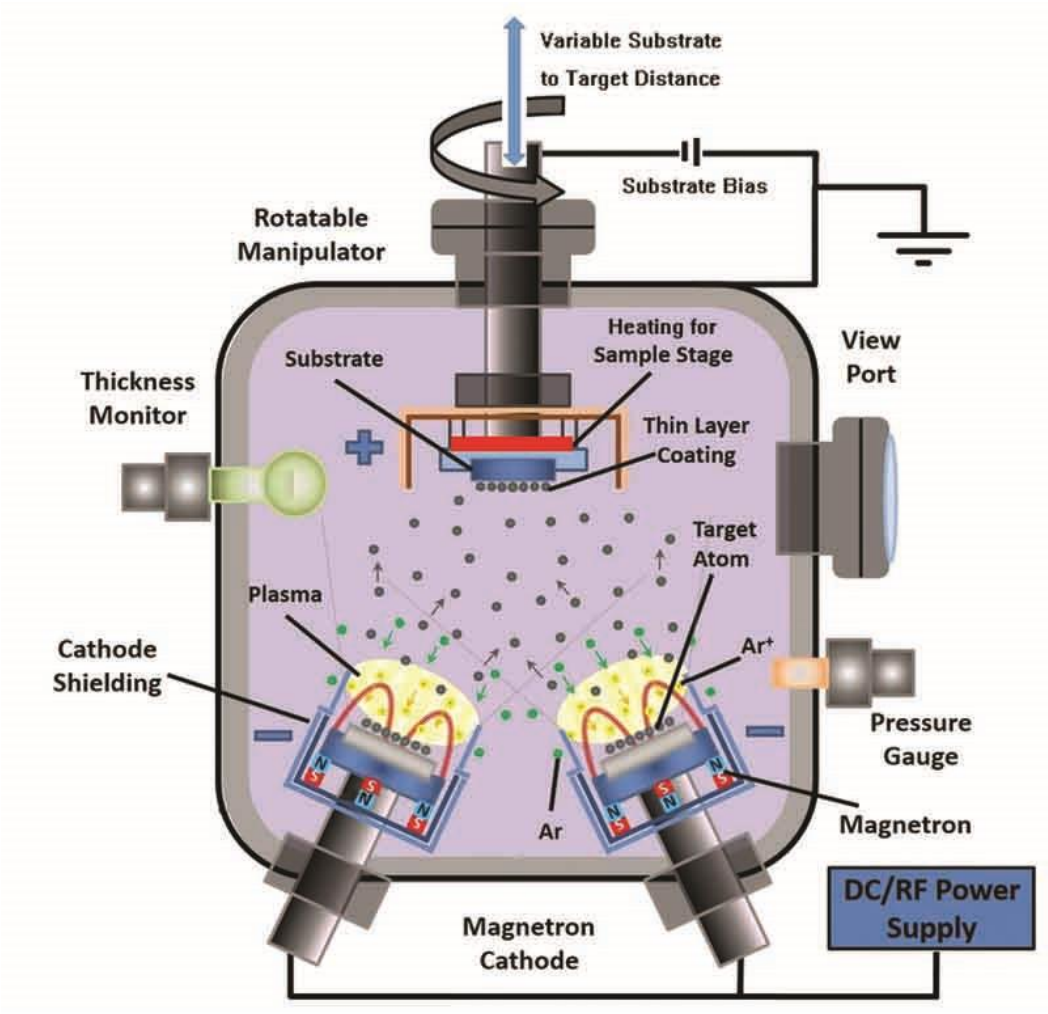
→ 薄膜制备是芯片加工中的一个重要工艺步骤

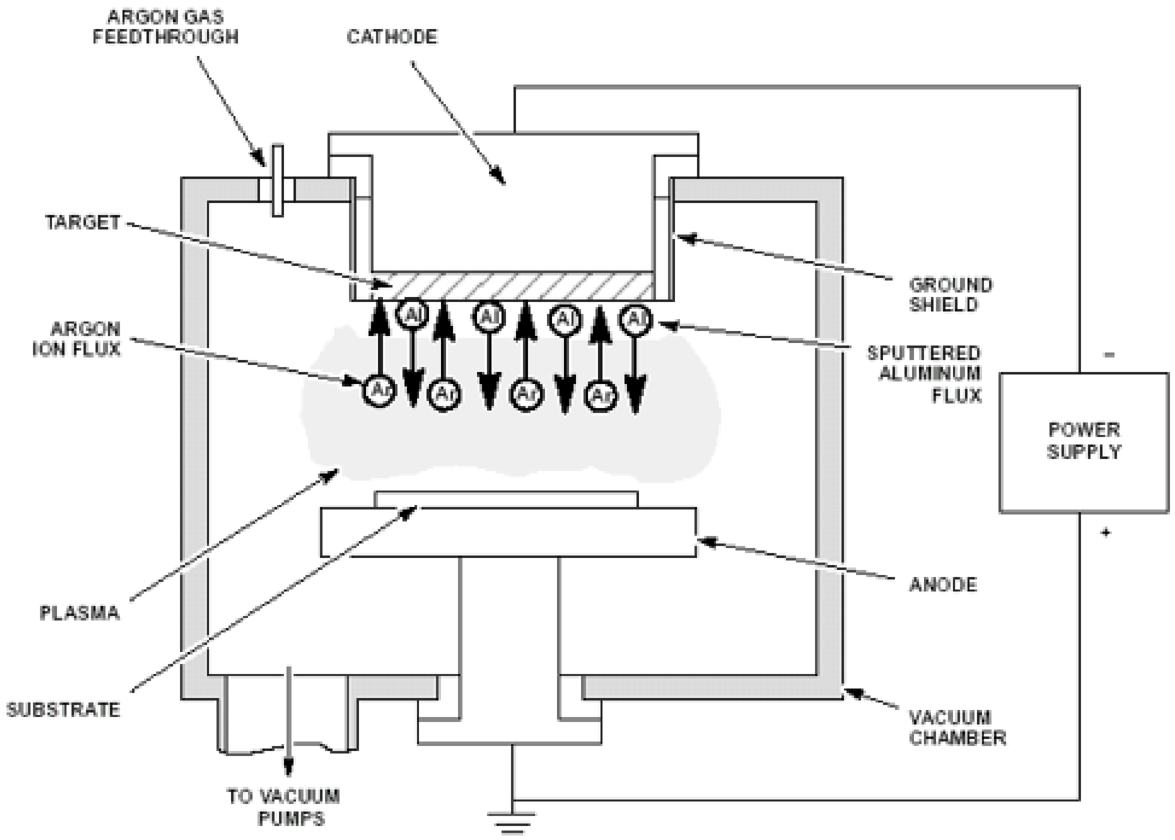
Outline

Equipments

- Sputtering is classified as a physical vapor deposition (PVD) process

Principles of Sputtering





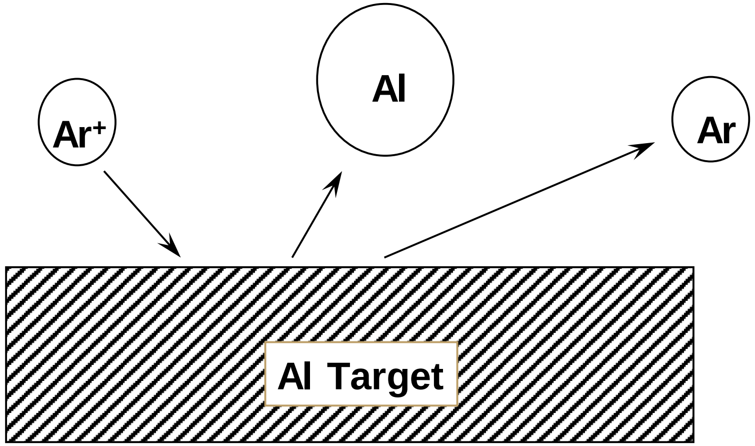
PVD Defined

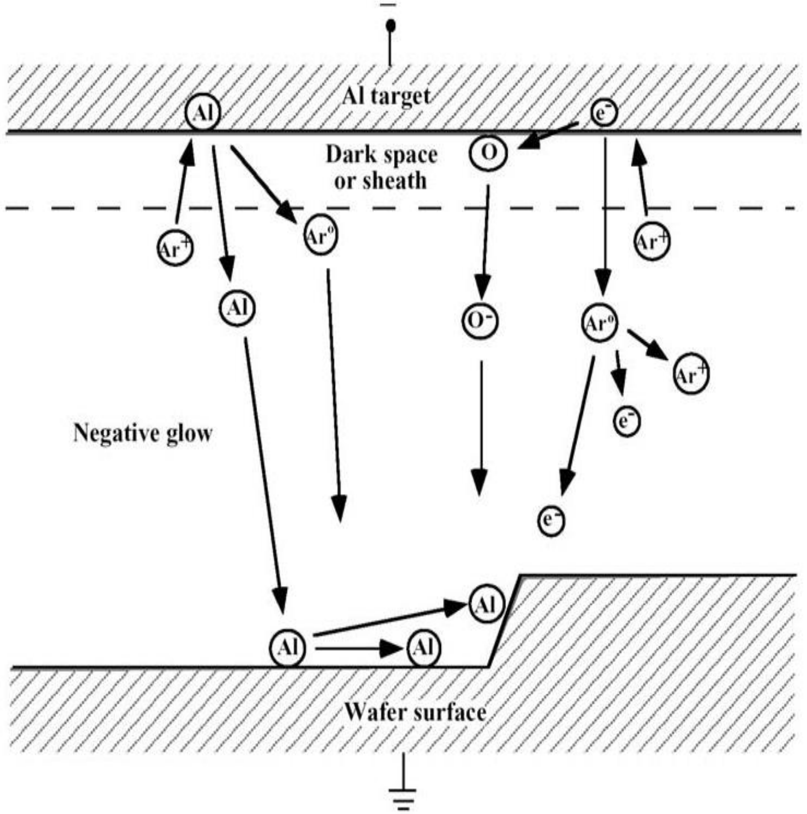
All PVD processes proceed according to the following sequence of steps:

- The material to be deposited is physically converted to a vapor phase.
- The vapor is transported across a region of reduced pressure (from the source to the substrate).
- The vapor condenses on the substrate to form a thin film.

What is Sputtering?

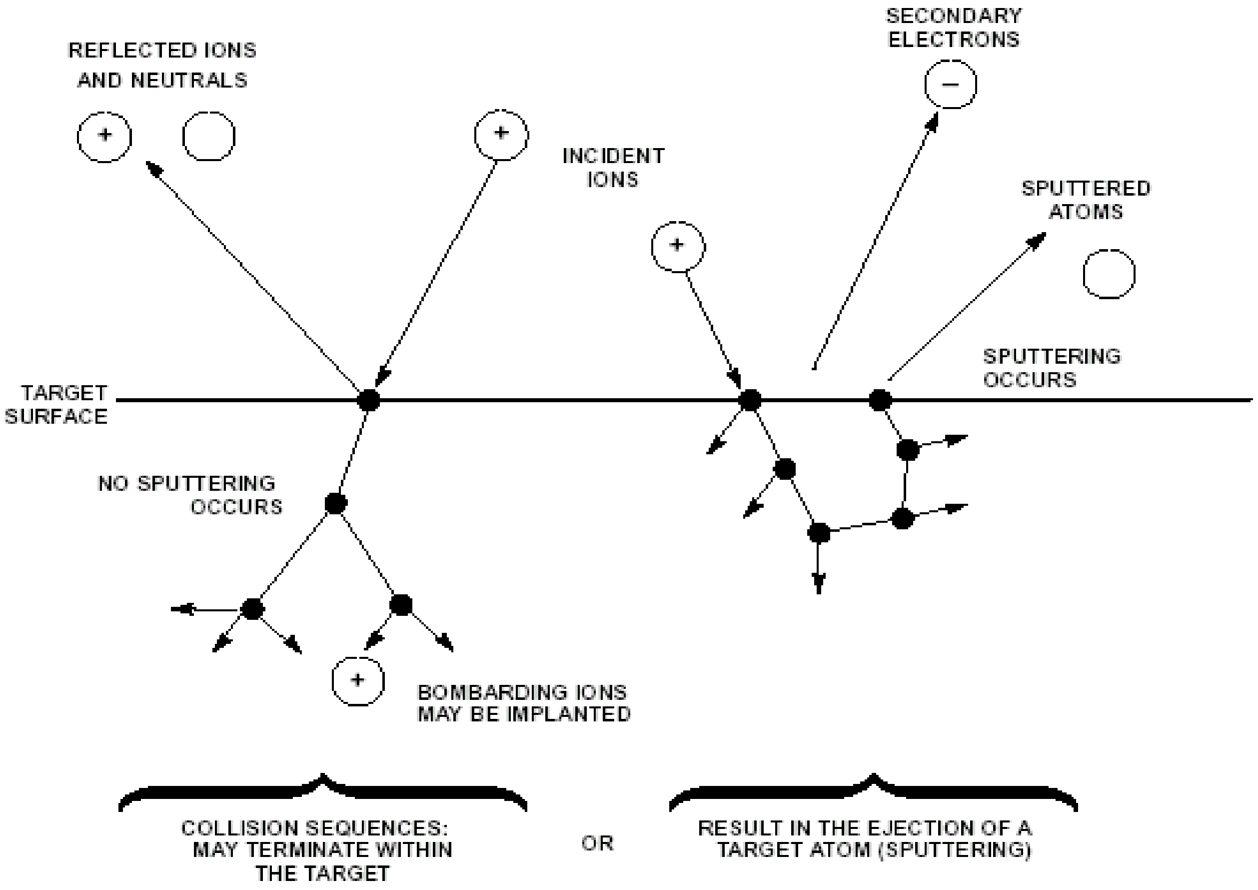
Sputtering - a process where atoms are removed from the surface of the target by the impact of energetic particles.



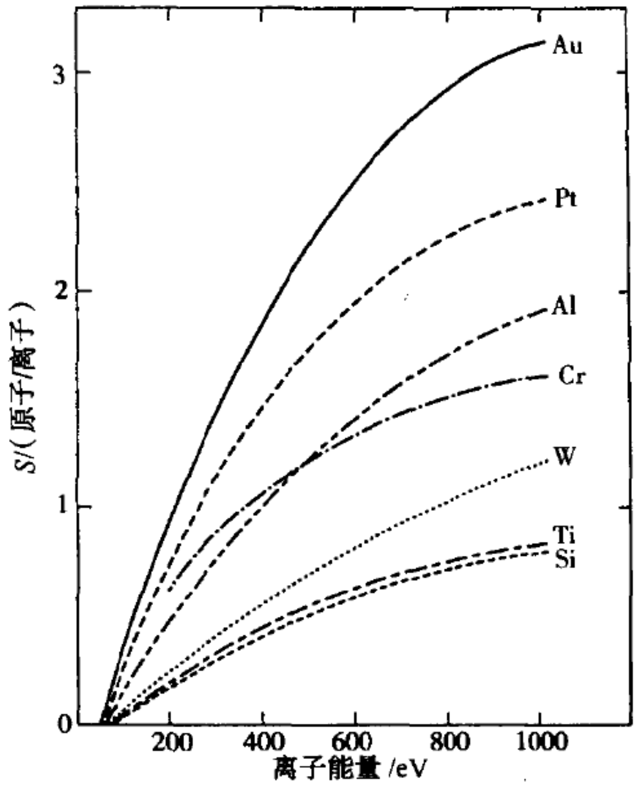


Principles of Sputtering

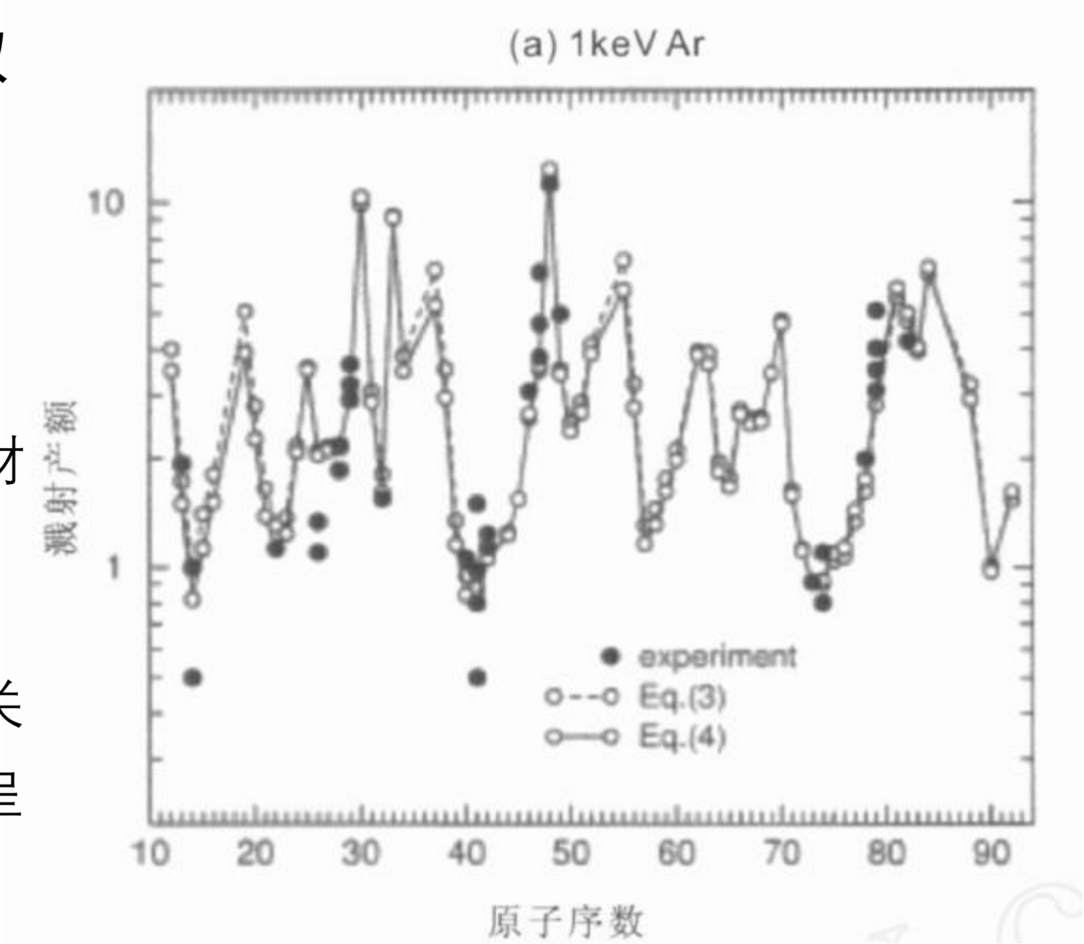
- Ions are generated and directed at a target (film source material).
- Film source atoms are dislodged from the surface of the target by collisions with the high energy ions.
- The ejected (sputtered) atoms are transferred to the substrate.
- The sputtered atoms impact the substrate where they condense and form a thin film.



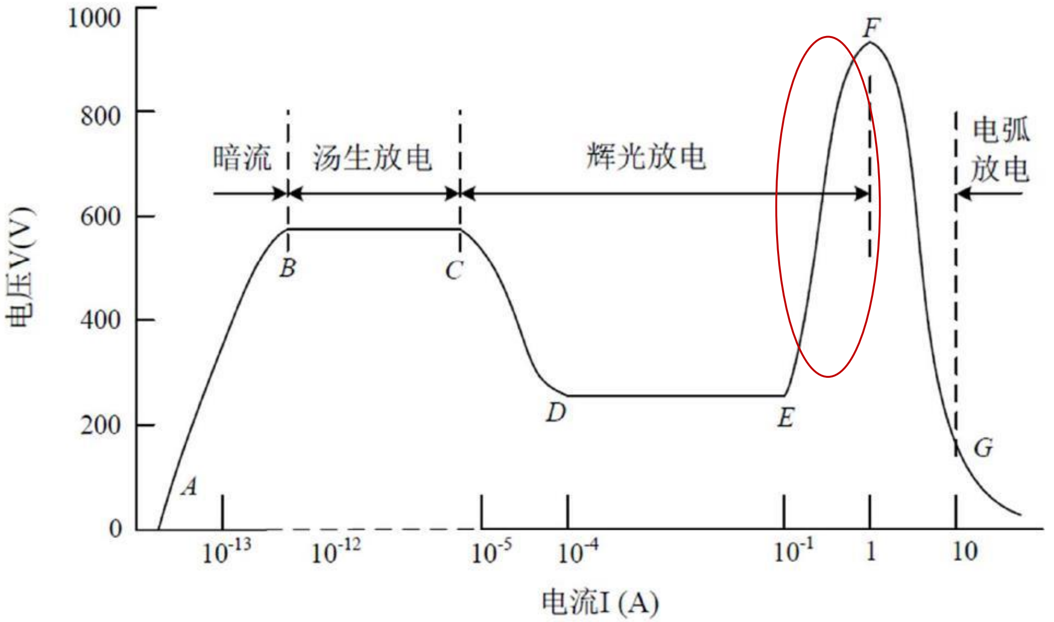
Principles of Sputtering

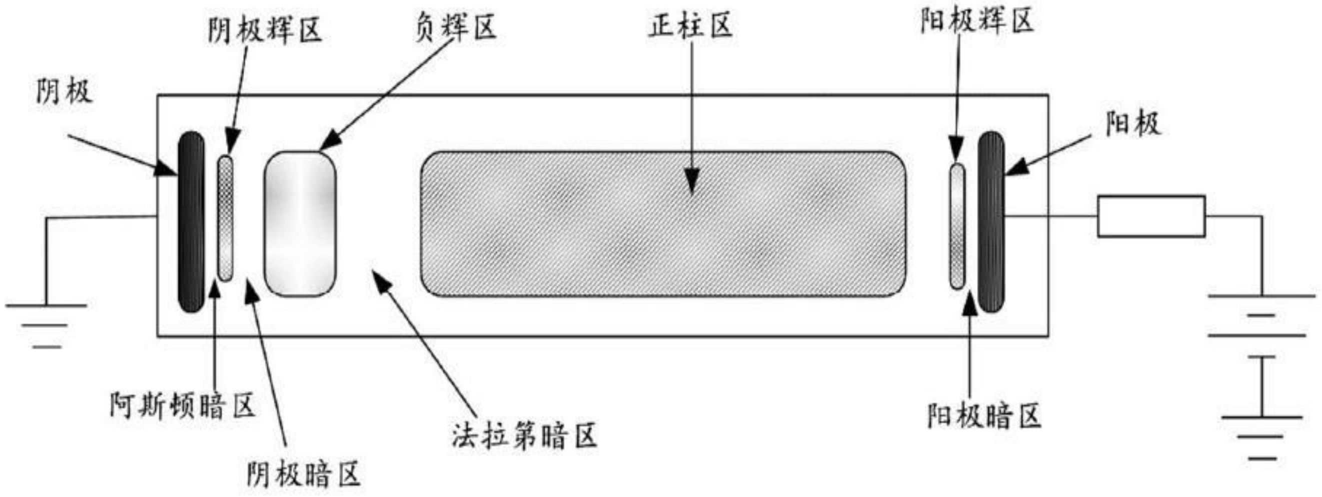


- 溅射阈值（10-30eV，取决于靶材材料种类）
- 溅射产额（溅射率：原子数比入射离子数）
  - 平均每个入射离子到靶材上，能打出的原子数
  - 与入射离子能量、种类、入射角以及靶材材料有关
  - 溅射率随靶材原子序数呈周期性变化



气体辉光放电-DC sputter



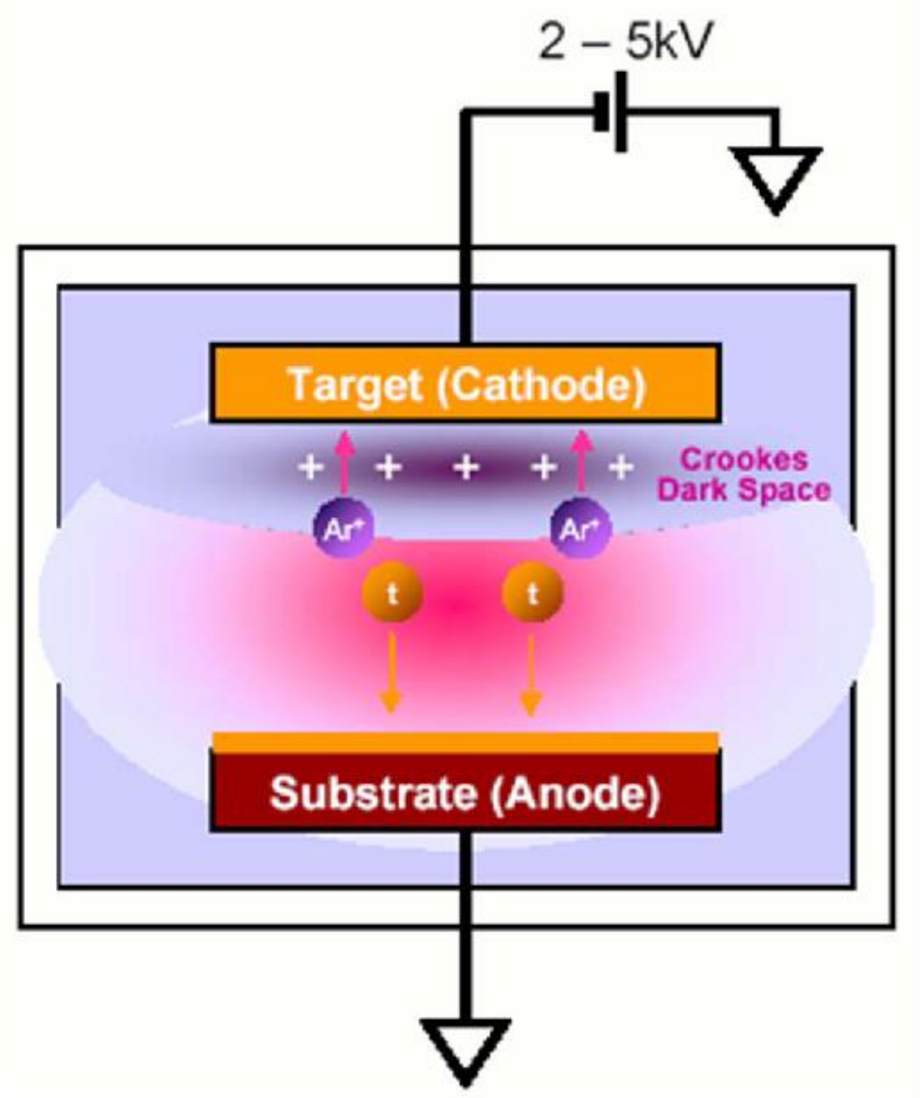


DE段，辉光放电，自持放电段，电流增加，放电电压电压低且不变
EF段，反常辉光放电段（**溅射区**），电流增加时放电电压升高，
辉光已占占据整个阴极

典型的直流辉光放电的区间分布

溅射分类-DC Sputter Deposition

- Free e- collide with Ar atoms
  - excitation of Ar atoms
  - ionization of Ar + secondary e-
- e- generate additional plasma
  Ar+ accelerated toward target
  - sputtering target material
- Deposition is not directional
  (unless collimated)



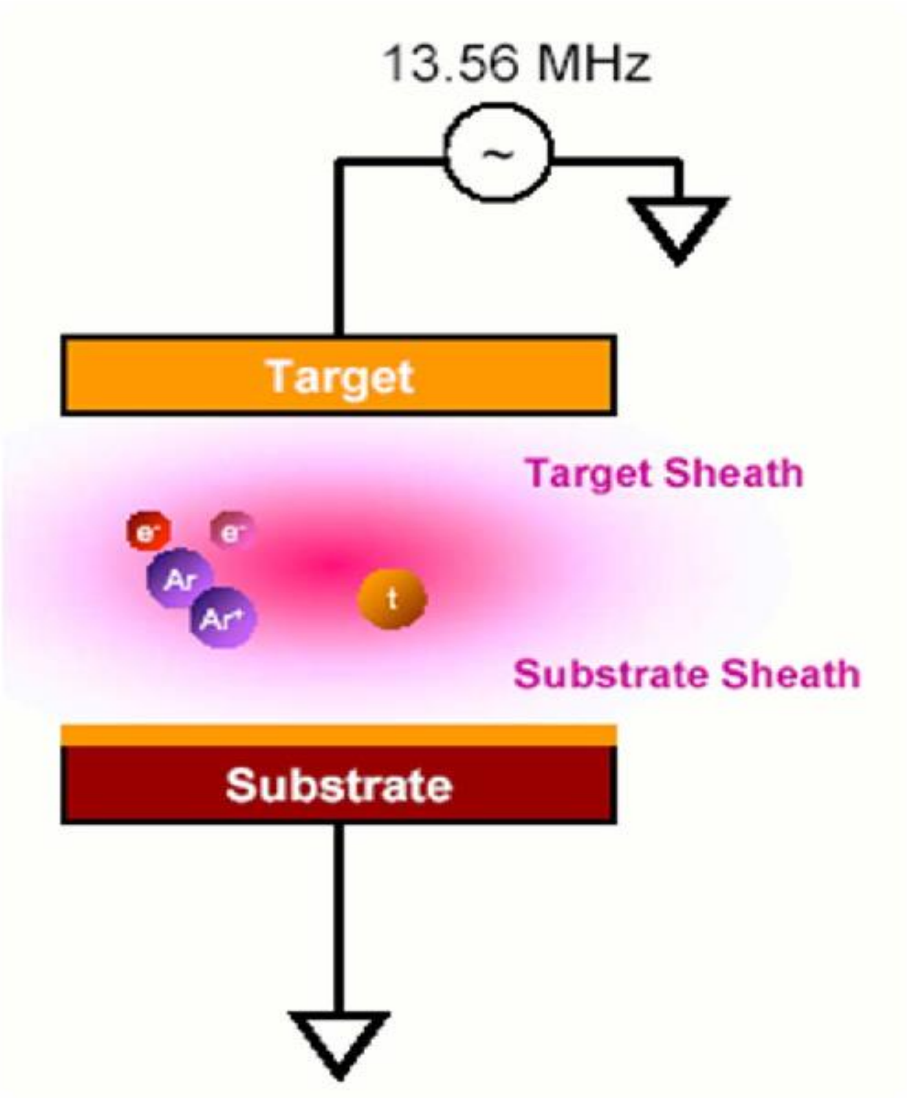
溅射分类-RF Sputtering

- RF field neutralizes positive charge buildup on electrodes
  - Allows deposition of dielectrics

- At low f (< 100 KHz), electrons and ions follow potential switching
- At high f (> 1 MHz), heavy ions cannot follow switching voltage

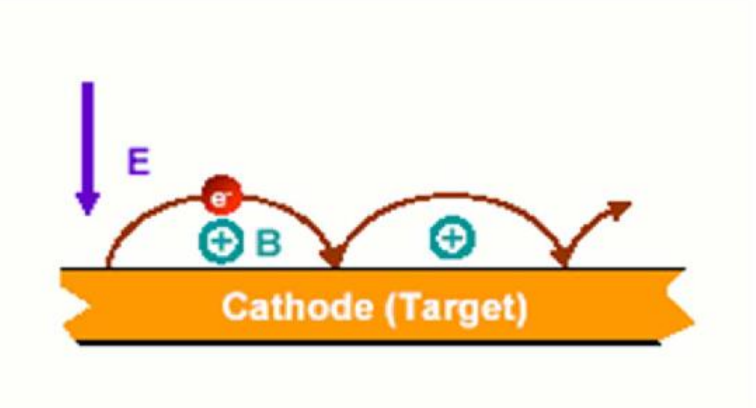
射频电场放电的特点:

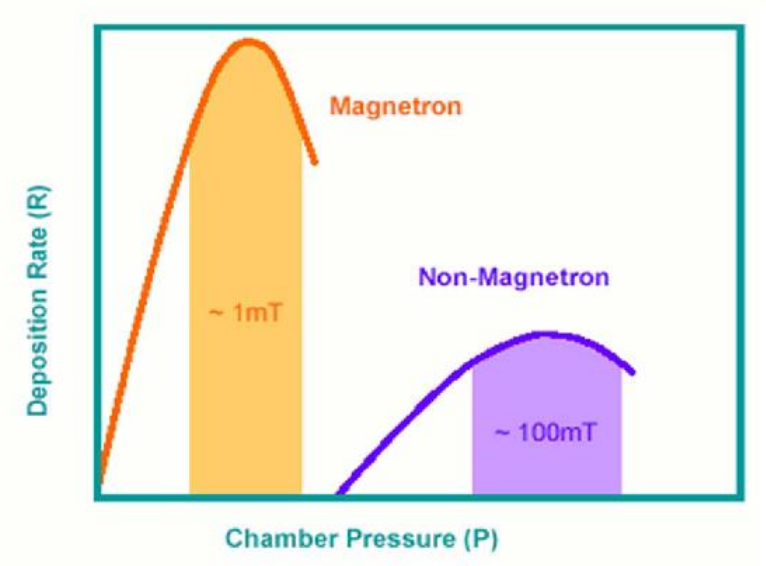
- 电子在两极间震荡运动，使气体容易电离，放电电压低
- 电极：导体或绝缘体



Magnetron Sputtering

- Magnetic field confines electrons to target area
- Current density increases
- Discharge pressure decreases
- Deposition rate increases

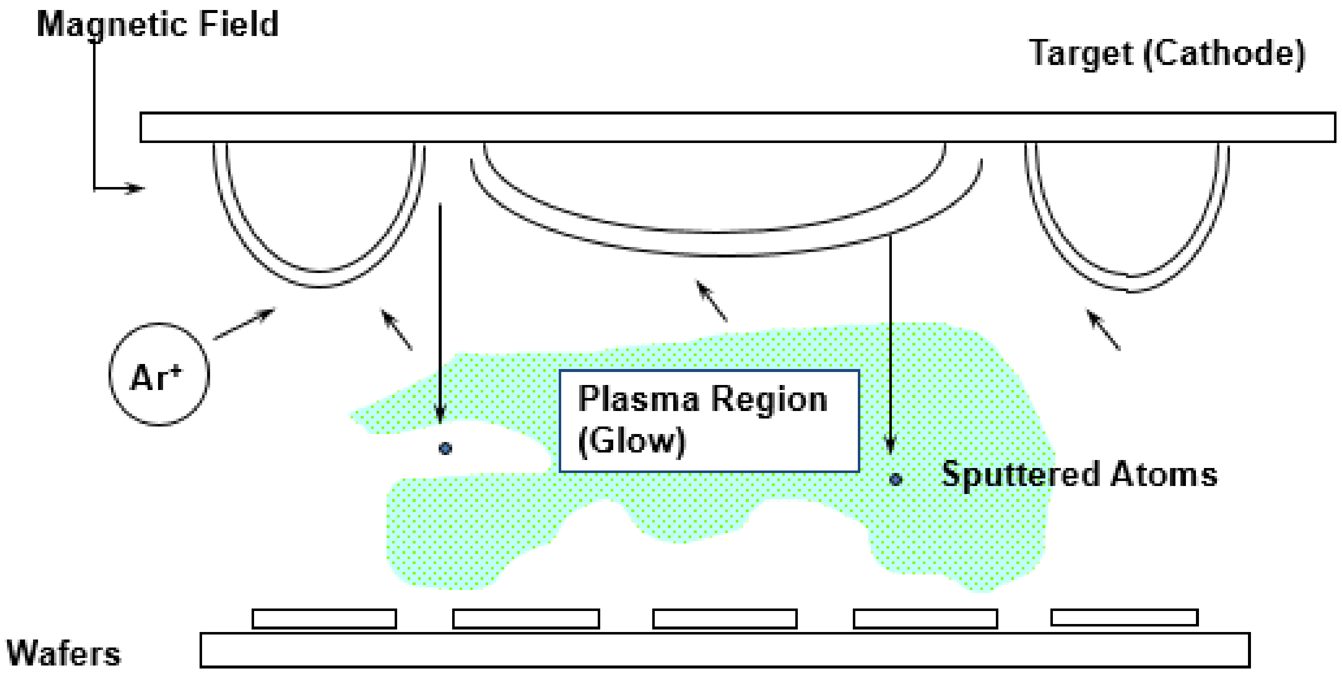




Magnetron Sputtering

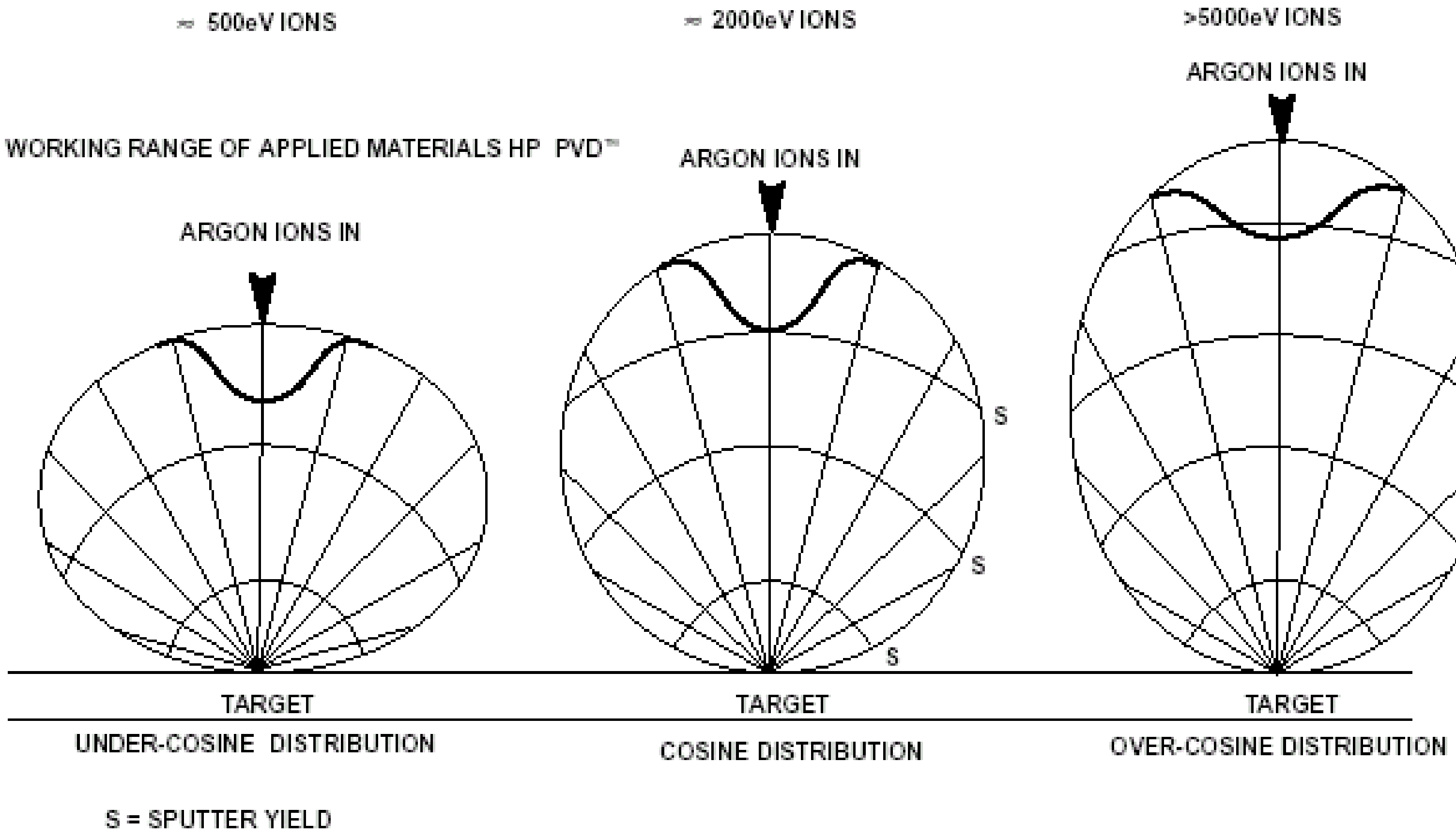
- Magnetic field can be designed to provide the erosion needed to obtain the proper uniformities.

- Typical pressure 1-20 mT.



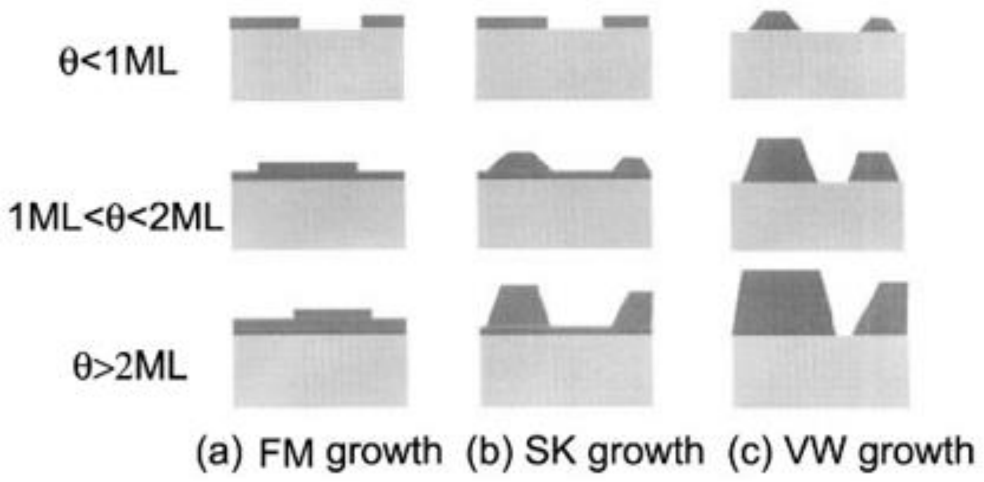
Key Challenge: Arrange magnets to obtain full surface erosion.

Sputtering Distribution



THESE GRAPHS SHOW THE ANGLES AT WHICH SPUTTERED ATOMS LEAVE THE SURFACE OF THE TARGET AS A FUNCTION OF INCIDENT ION ENERGY. ALL INCIDENT IONS ARE TRAVELLING PARALLEL TO THE SURFACE NORMAL OF THE TARGET (STRAIGHT INTO THE TARGET). THE HIGHER THE INCIDENT ION ENERGY, THE HIGHER THE ANGLES AT WHICH SPUTTERED ATOMS LEAVE THE TARGET SURFACE. AT LOWER ION ENERGIES THE ATOMS LEAVE THE SURFACE AT LOWER ANGLES.

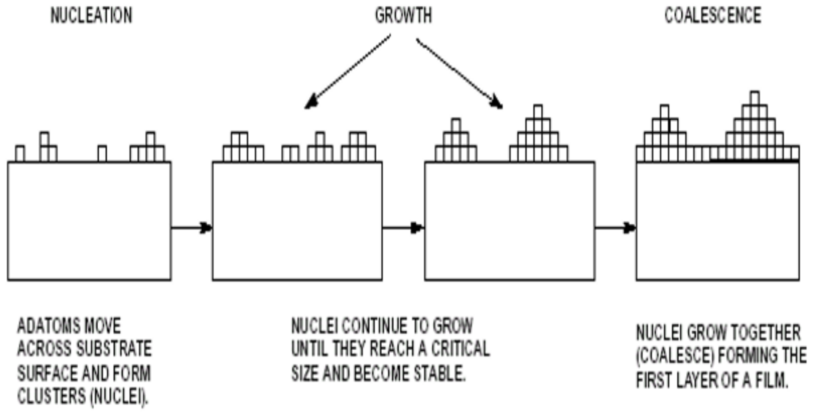
Nucleation and Grain Growth



Frank-van de Merwe (FM)

Stranski-Krastanov (SK)

Vollmer-Weber(VW)

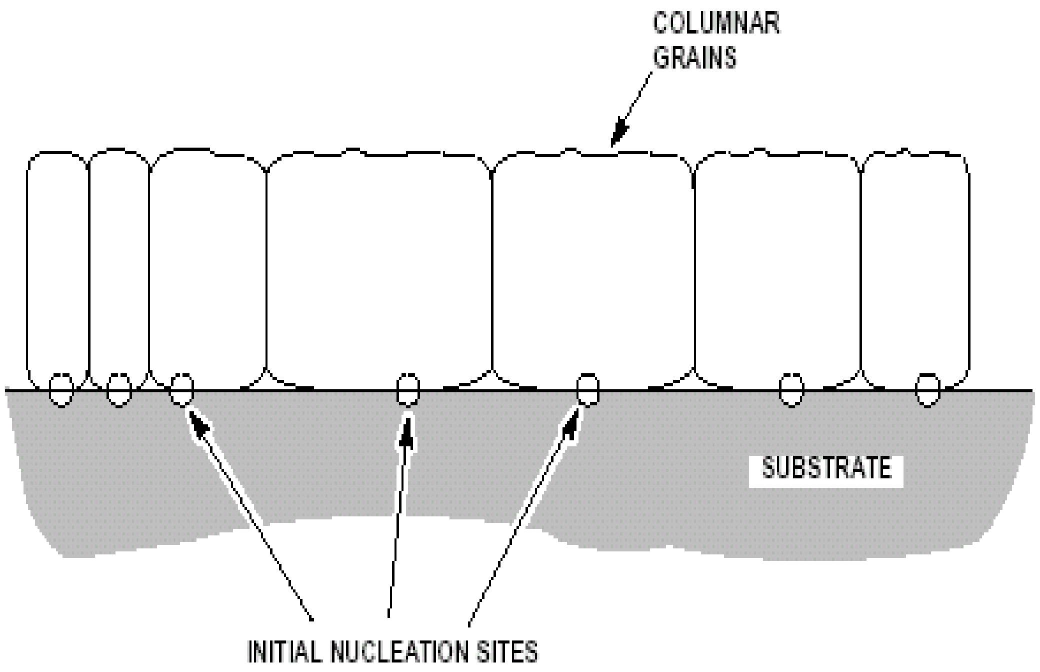


(a) Well matched between substrate and film $\mu_{sf} > \mu_{ss}$

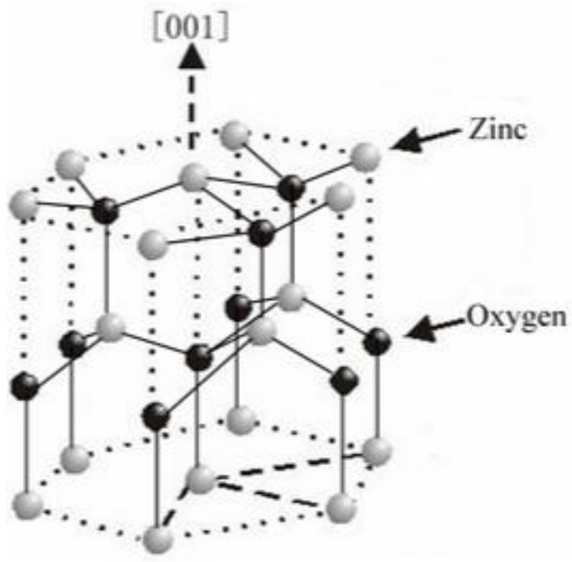
(b) Between a and c

(c) Large mismatch between sub. and film $\mu_{sf} < \mu_{ss}$

Nucleation and Grain Growth

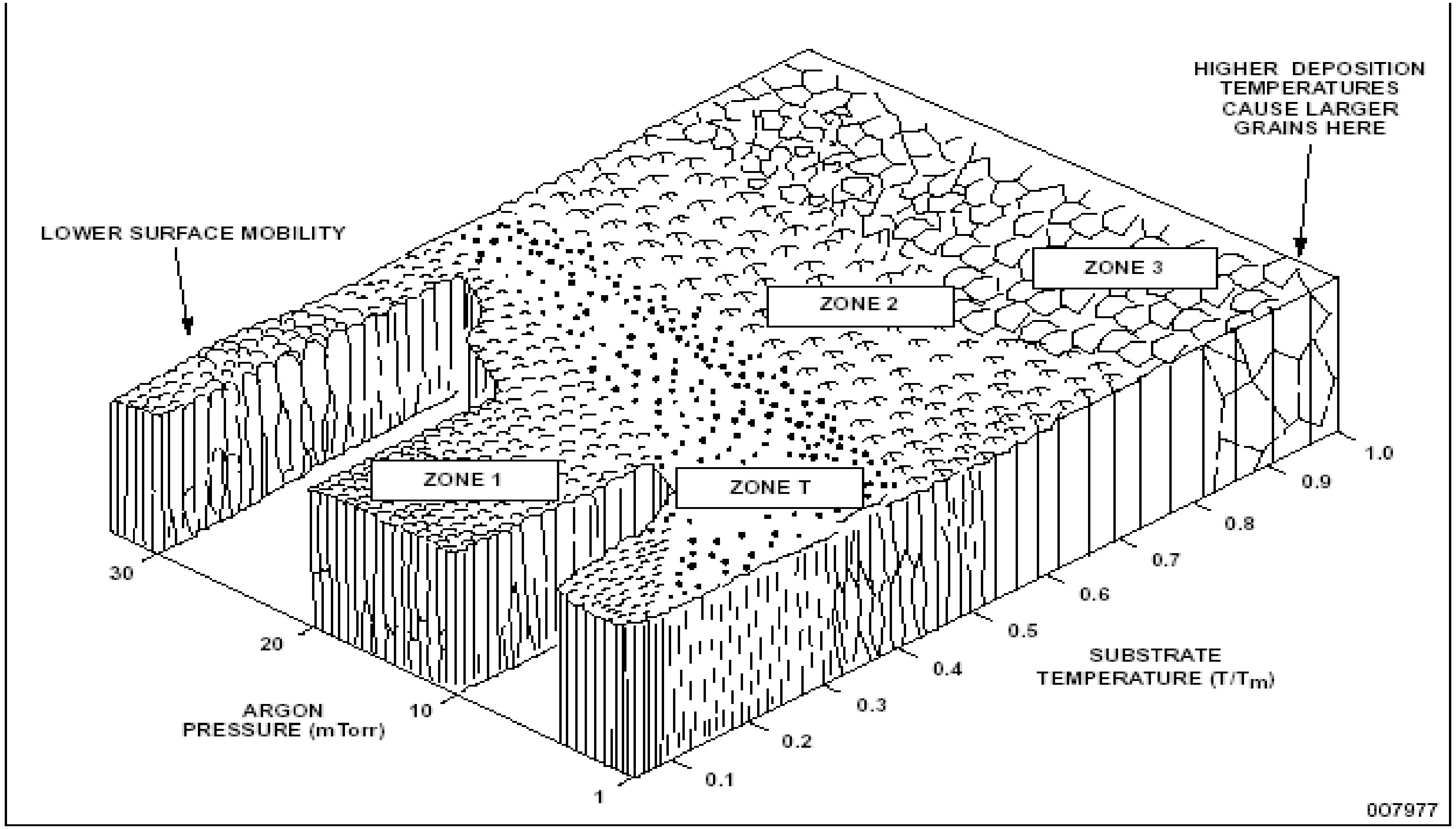


Columnar Grain Growth in a Polycrystalline Film

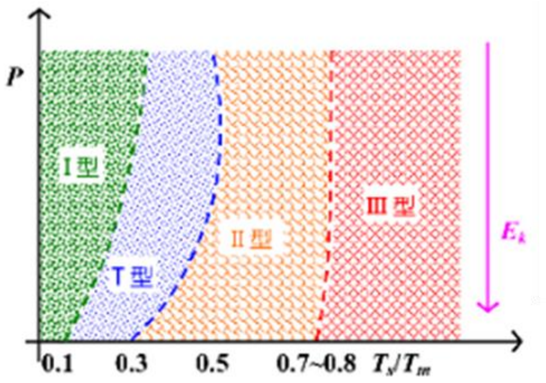


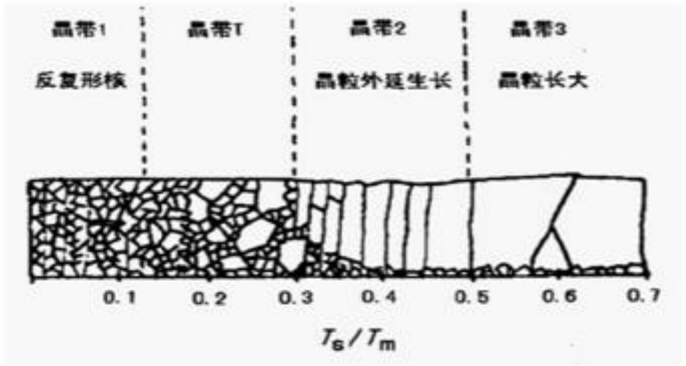
ZnSe thin film

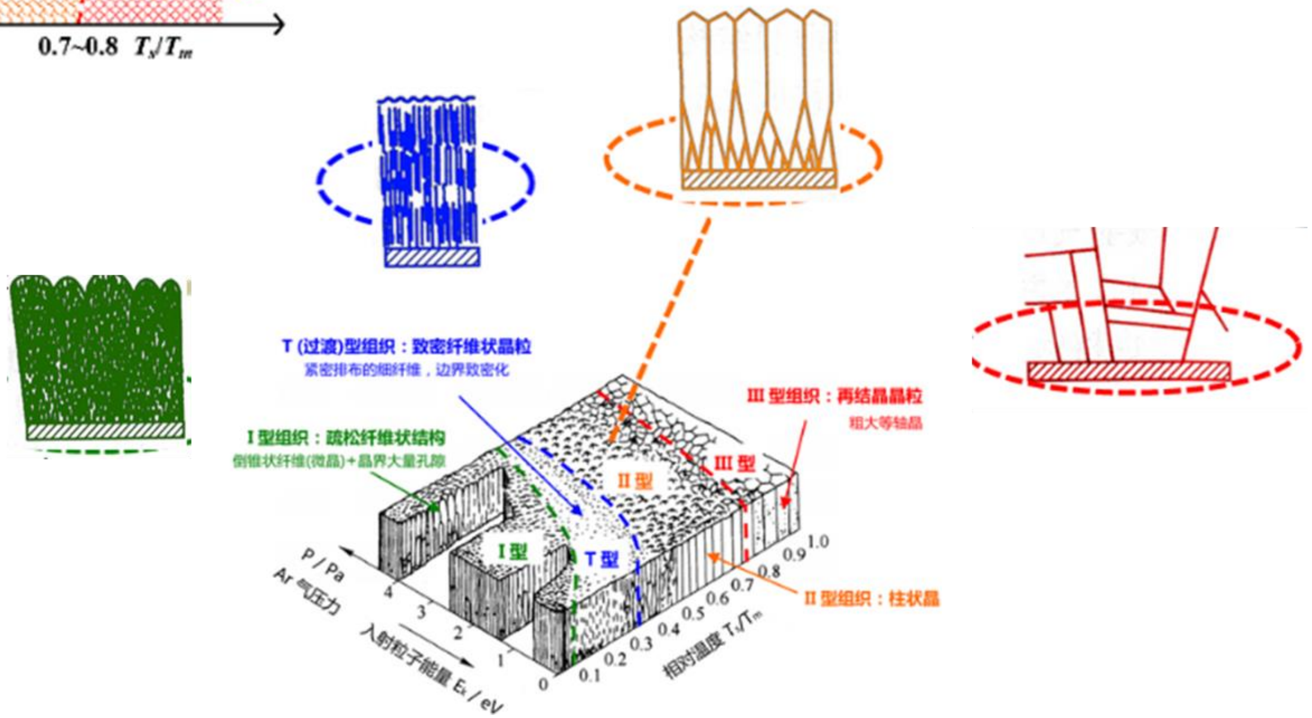
The relationship between substrate temperature and film growth



Using the MD Zone Diagram







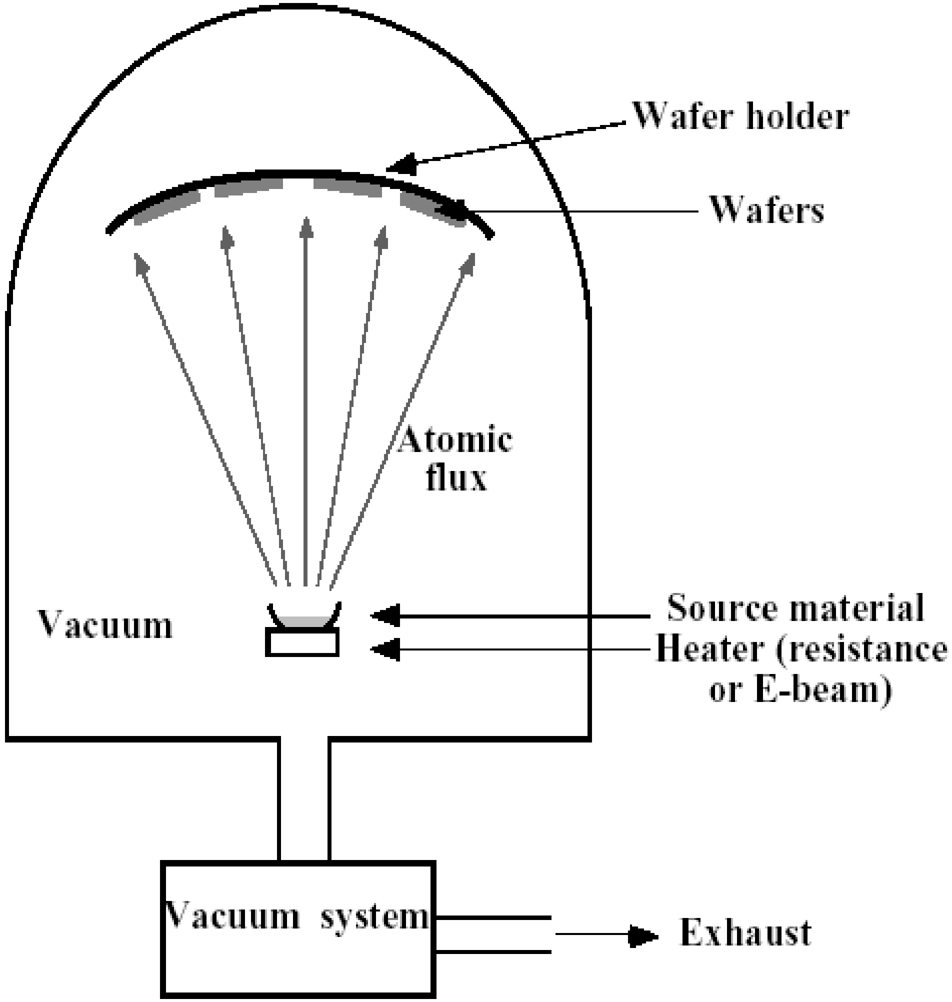
Outline

- 2. 1. Introduction
- 2. 2. Sputtering
- 2. 3. Evaporation
- 2. 4. Process Control
- 2. 5. Homework

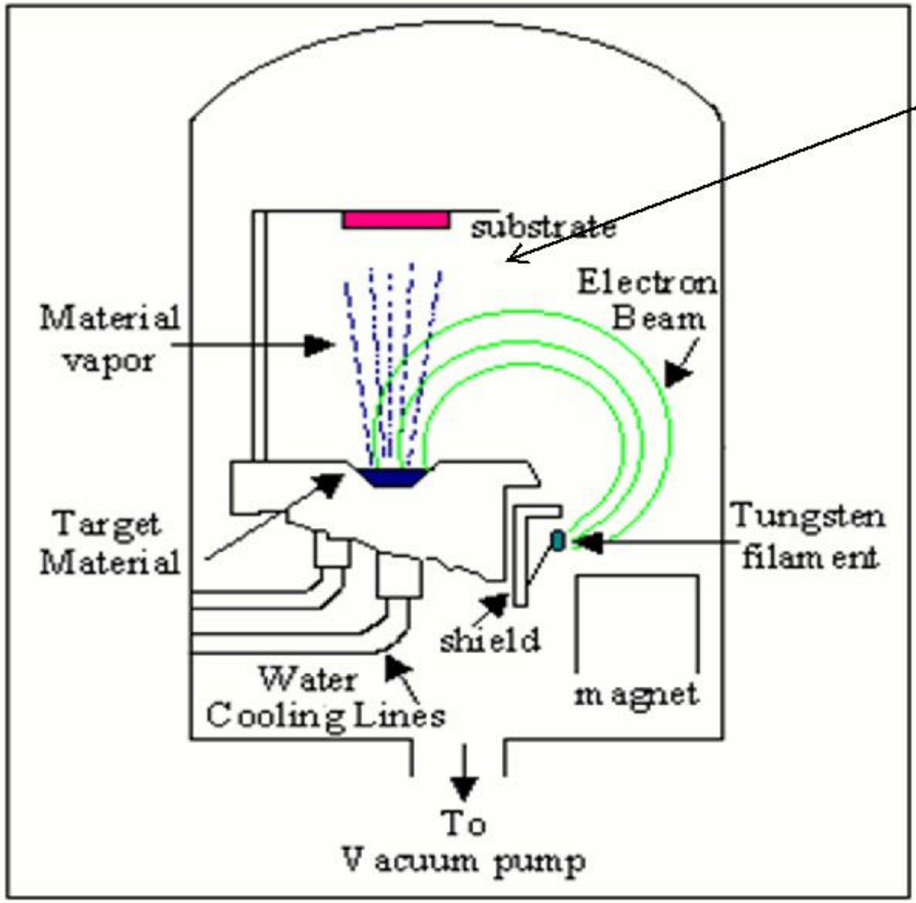
Thin film deposition

PVD: sputtering, evaporation

- Evaporation
  - – thermal (Ti, Al, Cu, TiN & TaN *et al.*)
  - – e-beam(高熔点材料)
  - -激光加热
  - -高频感应加热

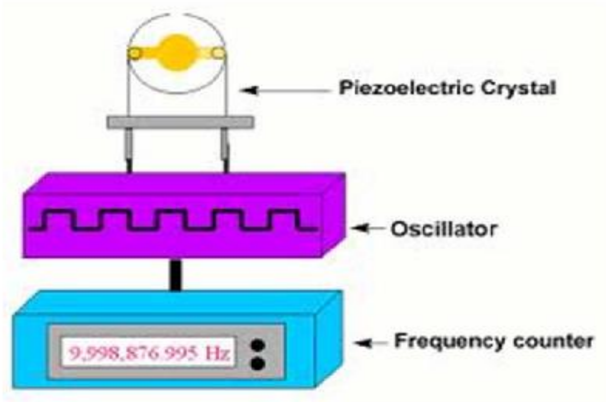


E-beam Evaporation

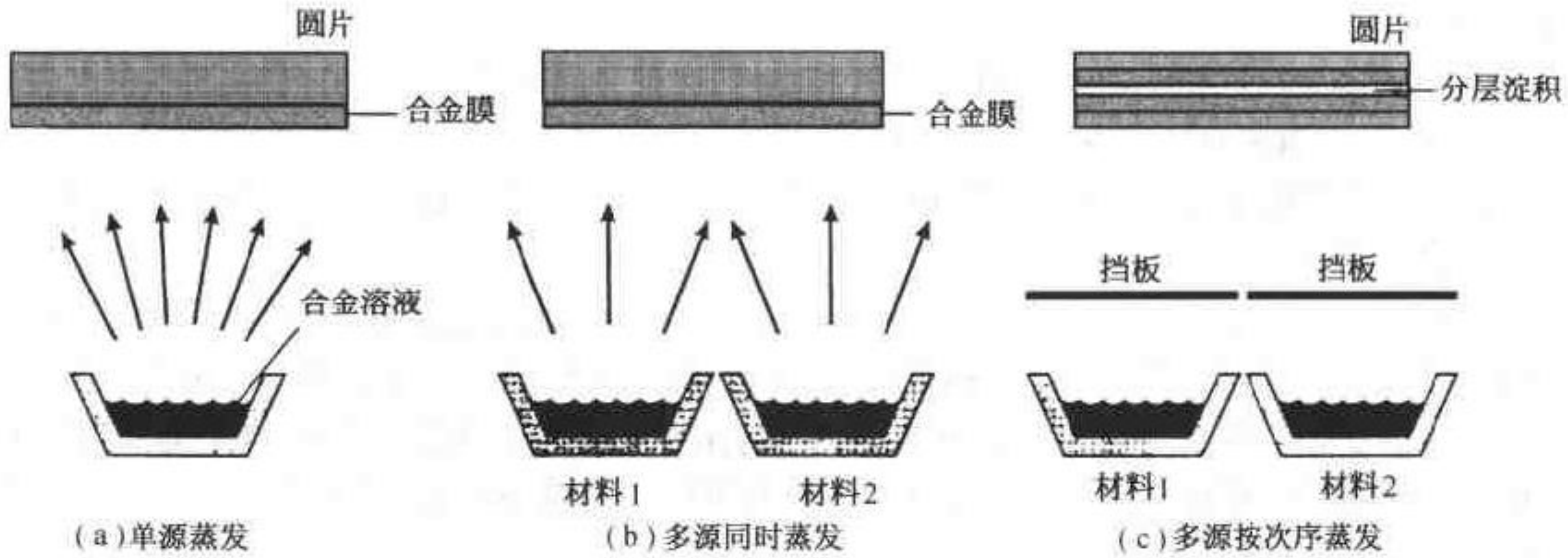


Evaporation system

Quartz crystal microbalance



Multiple Composition Evaporation



合金靶中各组分材料的蒸汽压接近

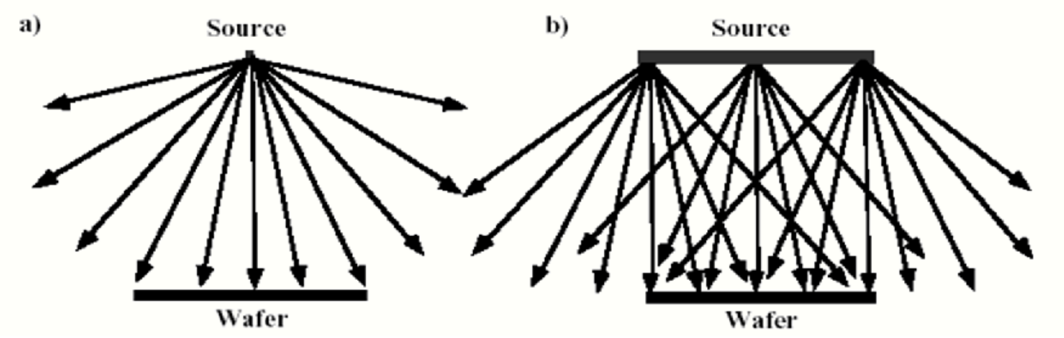
不同温度多源同时蒸发

蒸发结束后高温退火

Thermal vs. E-beam Evaporation

Deposition	Material	Typical Evaporant	Impurity	Deposition Rate	Temperature Range	Cost
Thermal	Metal or low melt-point materials	Au, Ag, Al, Cr, Sn, Sb, Ge, In, Mg, Ga CdS, PbS, CdSe, NaCl, KCl, AgCl, MgF_2, CaF_2, $PbCl_2$	High	1 ~ 20 A/s	~ 1800 ºC	Low
E-Beam	Both metal and dielectrics	Everything above, plus: Ni, Pt, Ir, Rh, Ti, V, Zr, W, Ta, Mo Al2O3, SiO, SiO2, SnO2, TiO2, ZrO2	Low	10 ~ 100 A/s	~ 3000 ºC	High

Evaporation vs. Sputtering



Evaporation	Sputtering
Low energy atoms (~ 0.1 eV)	High energy atoms / ions (1 – 10 eV) • denser film • smaller grain size • better adhesion
High Vacuum • directional, good for lift-off • lower impurity	Low Vacuum • poor directionality, better step coverage • gas atom implanted in the film
Point Source • poor uniformity	Parallel Plate Source • better uniformity
Component Evaporate at Different Rate • poor stoichiometry	All Component Sputtered with Similar Rate • maintain stoichiometry

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- 2. 1. Introduction
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- 2. 3. Evaporation
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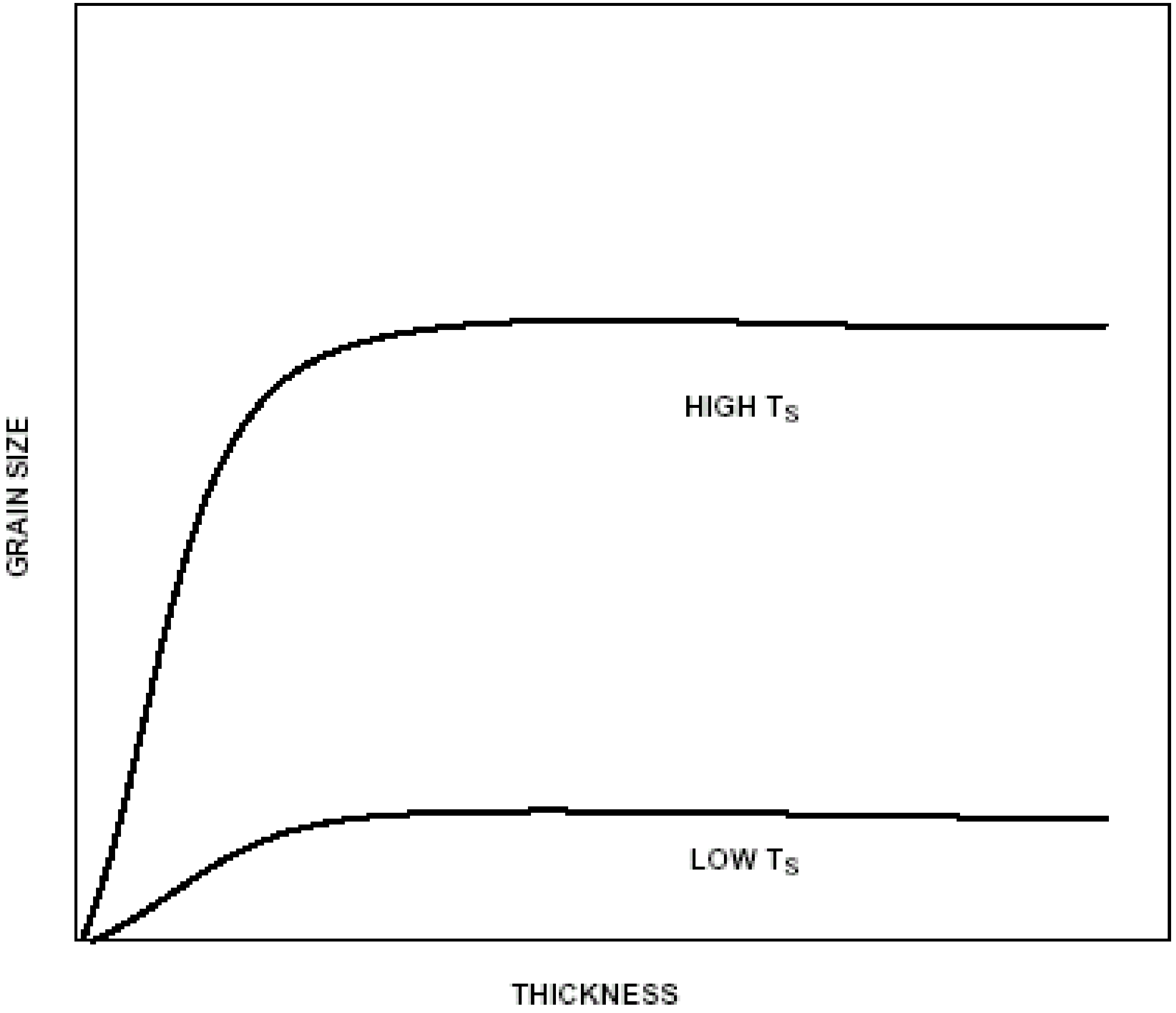
- 2. 5. Homework

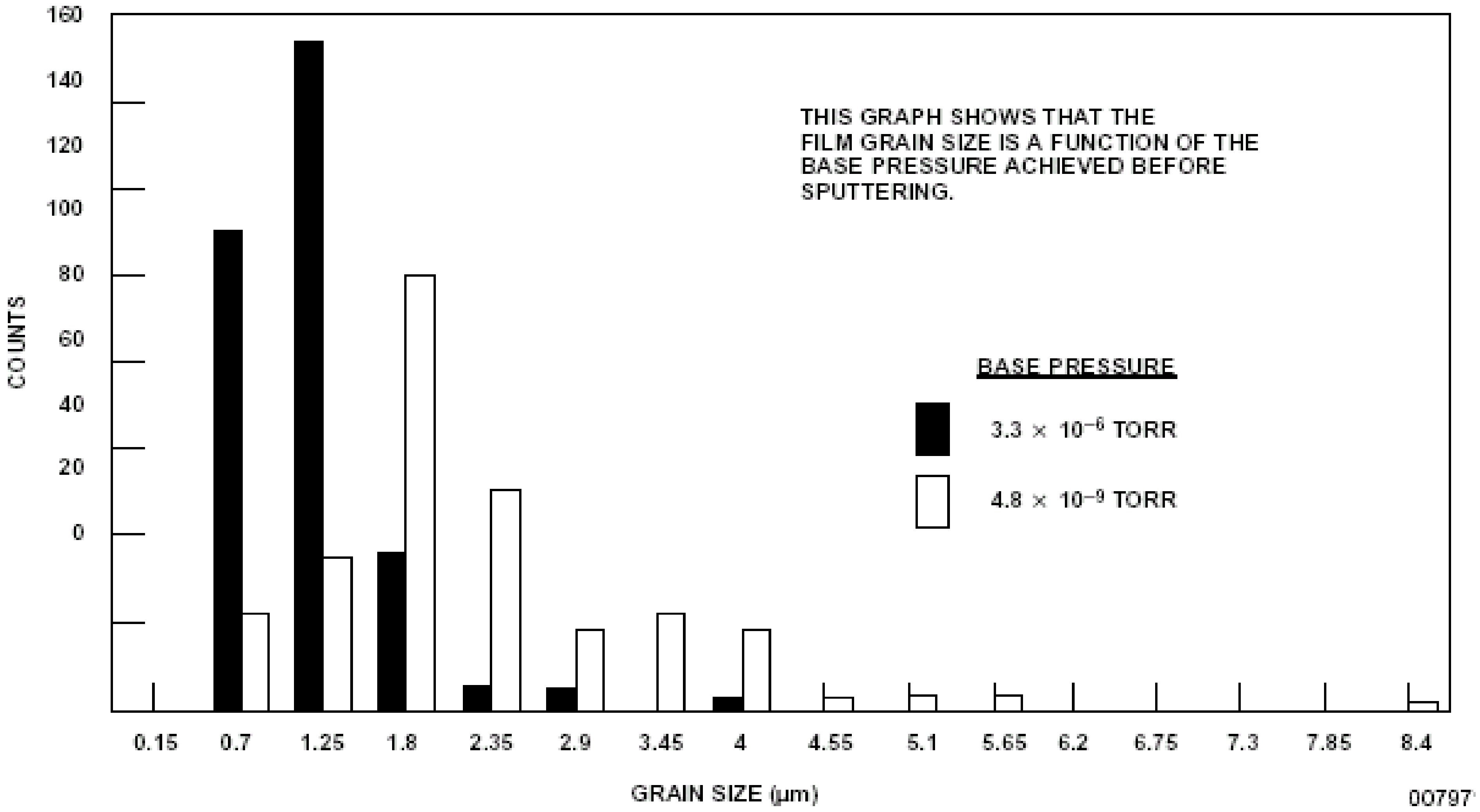
PVD Process Control

- Main process parameter
- Pressure
- Temperature
- Power
- Spacing

Variation of Film Grain Size with Substrate Temperature



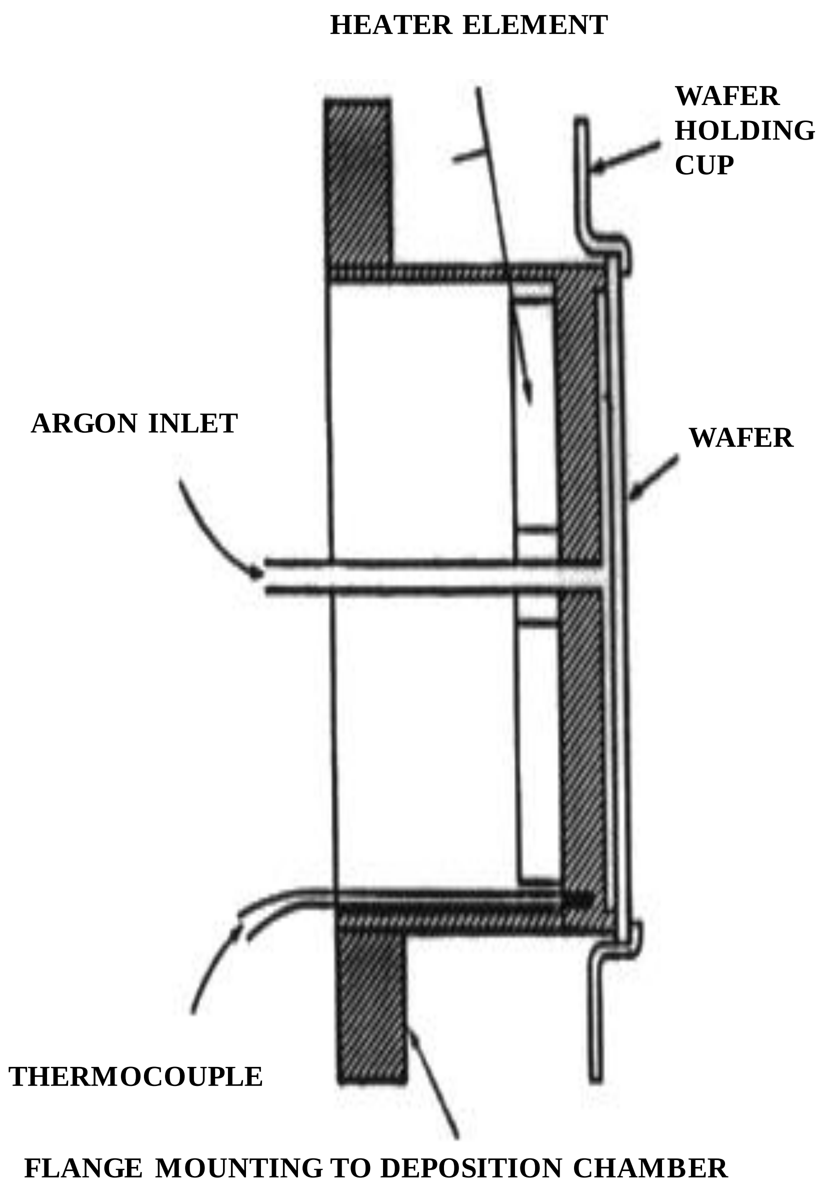
Variation of Film Grain Size with Substrate Surface cleanliness



Process Control

Main process parameter

- Wafer Temperature Control:
  - Wafer temperature must be precisely controlled during sputtering. The most common means of controlling wafer temperature is by using back-side heat conduction.
    - High pressure Argon is applied to the backside of the wafer to provide a thermodynamic coupling (via conductive heat transfer) to a heater chuck below the wafer.



Process Control

- Pressure control and measurement

1 atmosphere of pressure = 760 torr (1 torr = 1mm of mercury)

= 1 Bar

= 10E5 Pascals (N/m^2)

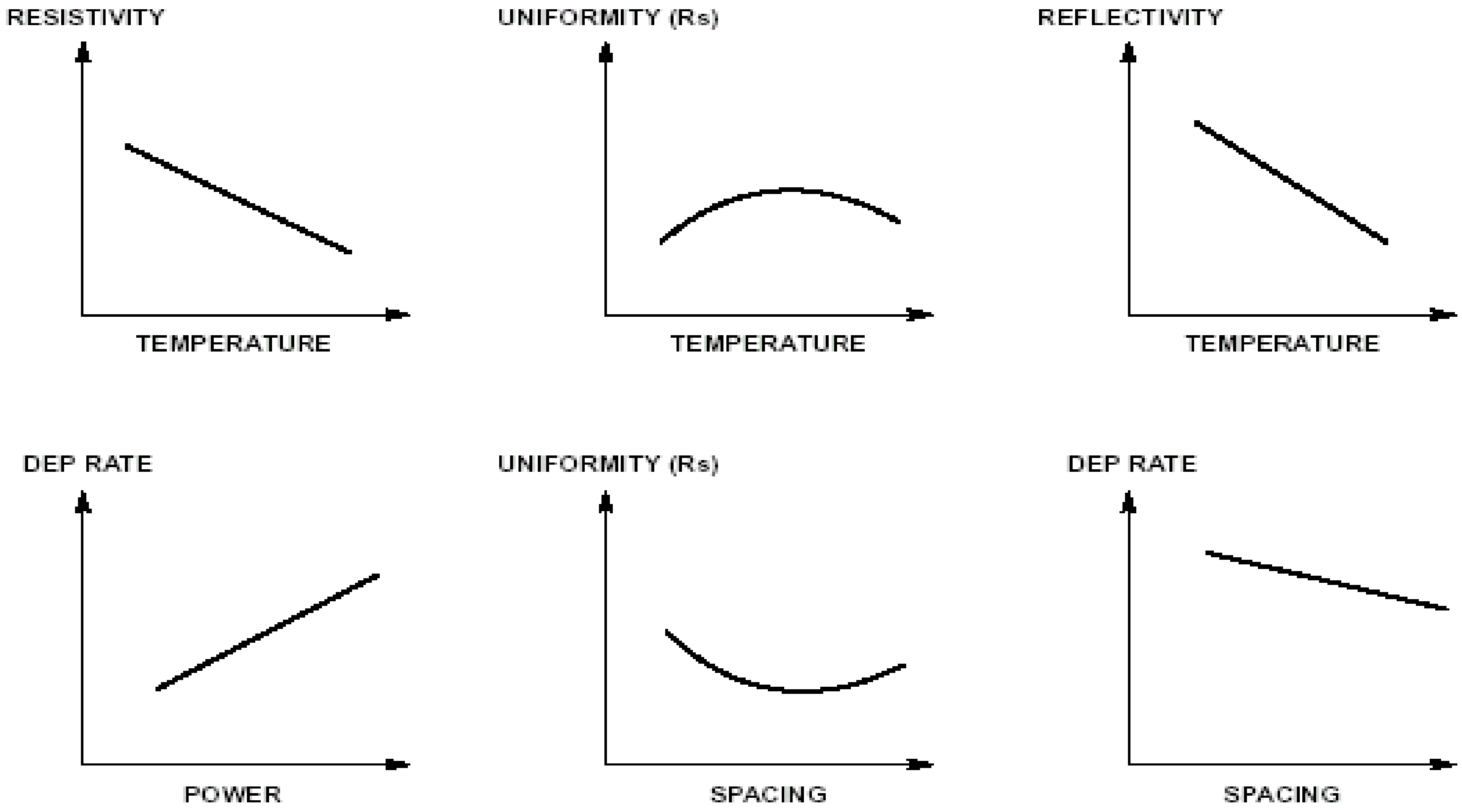
= 14.7 pounds/$inch^2$

Common Conversions: 1 mbar = 160 mtorr; 1 Pa = 7.6 mtorr

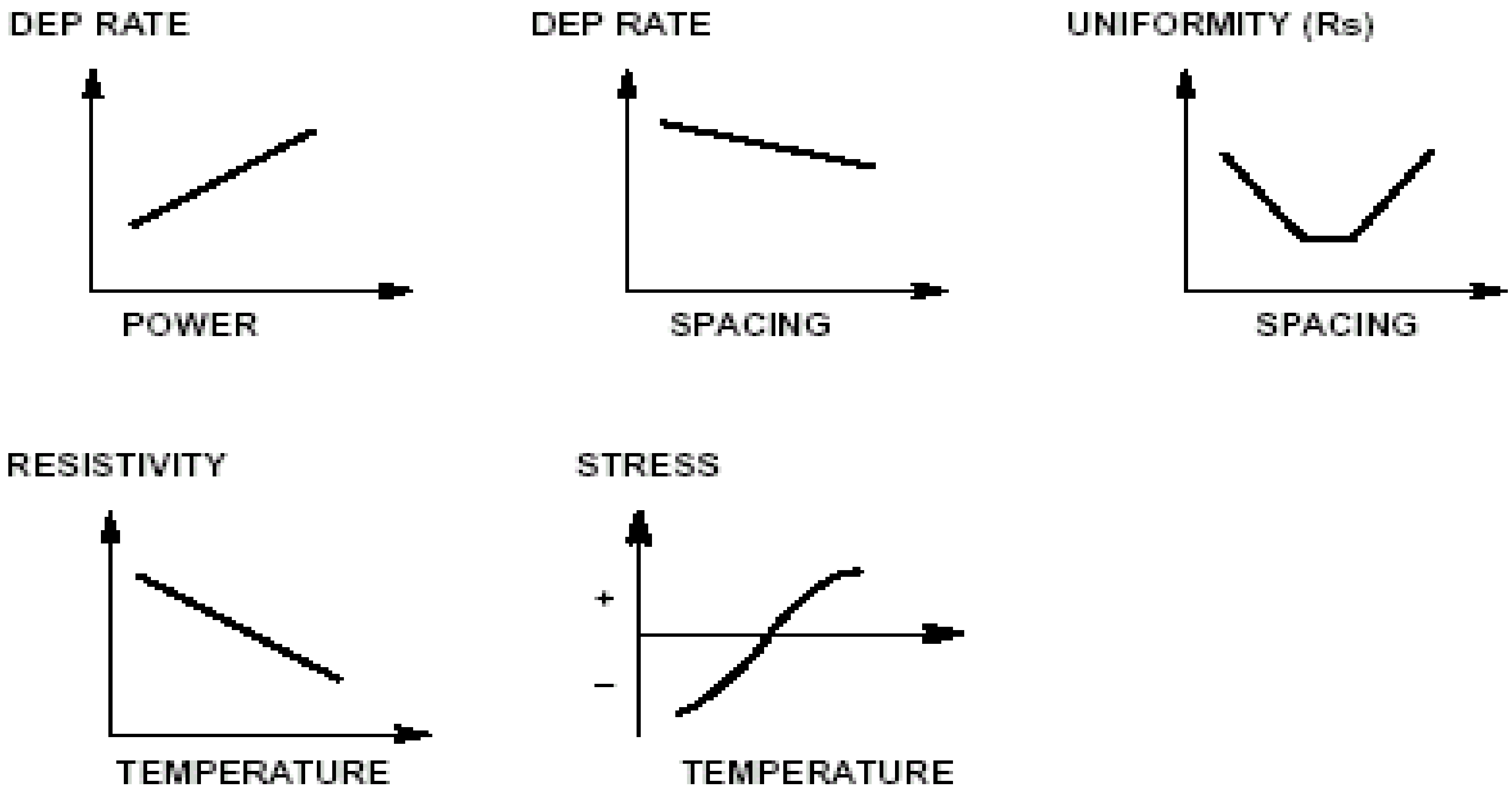
Pressure Gauges:

1. Convection gauge	**Atm. - 50mT**	**(rough pressures)**
2. Baratron	**100mT - 1mT**	**(process pressures)**
3. Ionization gauge (hot filament or cold cathode)	**100mT - 10E-9T**	**(base pressure)**

Al Trend Charts

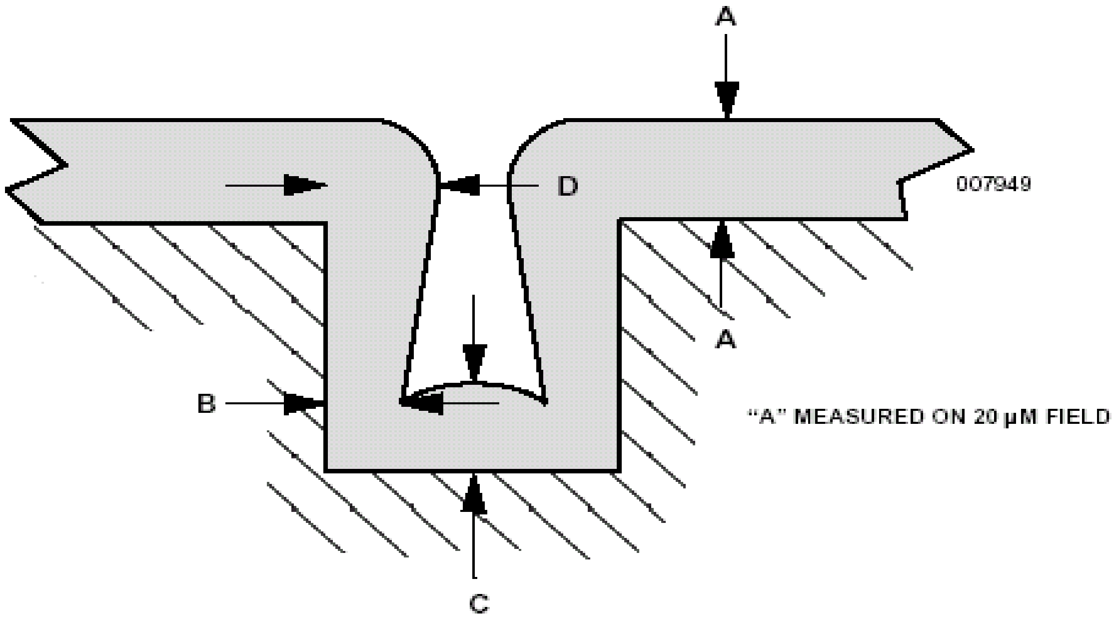


Ti Trend Charts



Thin Film Deposition in trench or via

Step Coverage Measurement Criteria



BOTTOM COVERAGE = (C/A) × 100

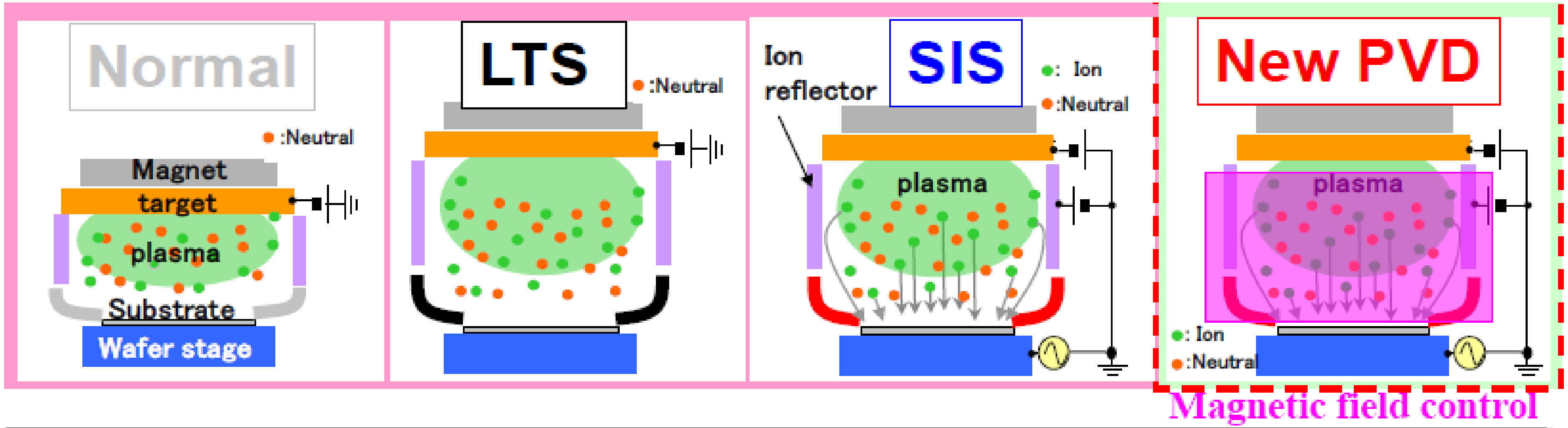
SIDE MINIMUM COVERAGE = (B/A) × 100

OVERHANG = ((D − B))/B × 100

Thin Film Deposition in trench or via

Coverage in trench-challenge

	2009	2010	2011	2012	2013
Via dia	$\phi 50\mu m$		$\phi 25\mu m$		$\phi 6\mu m$
A/R	<3		<3−6		<6−12
sputter	LTS or B-LTS		SIS		New PVD

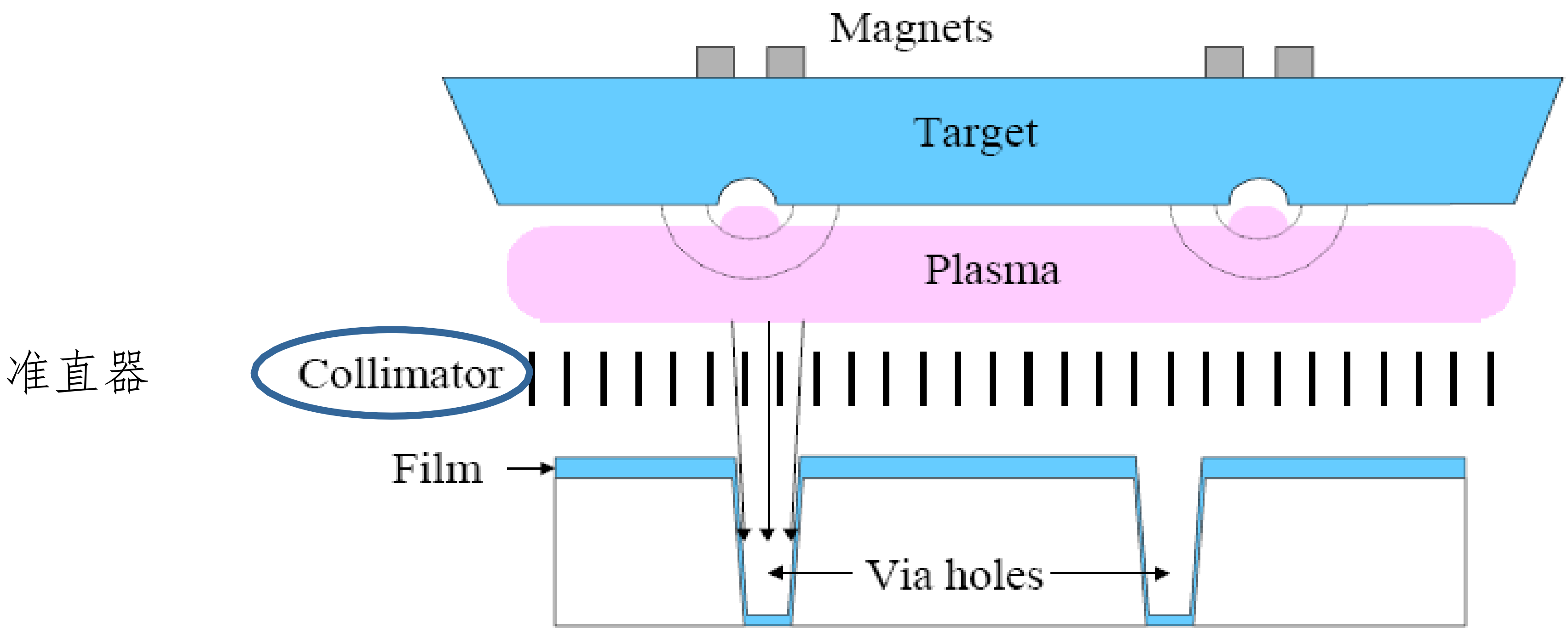


Depo	Normal	LTS	SIS(B-LTS)	Side Coil
AR	<2	2−3	4−6	6<

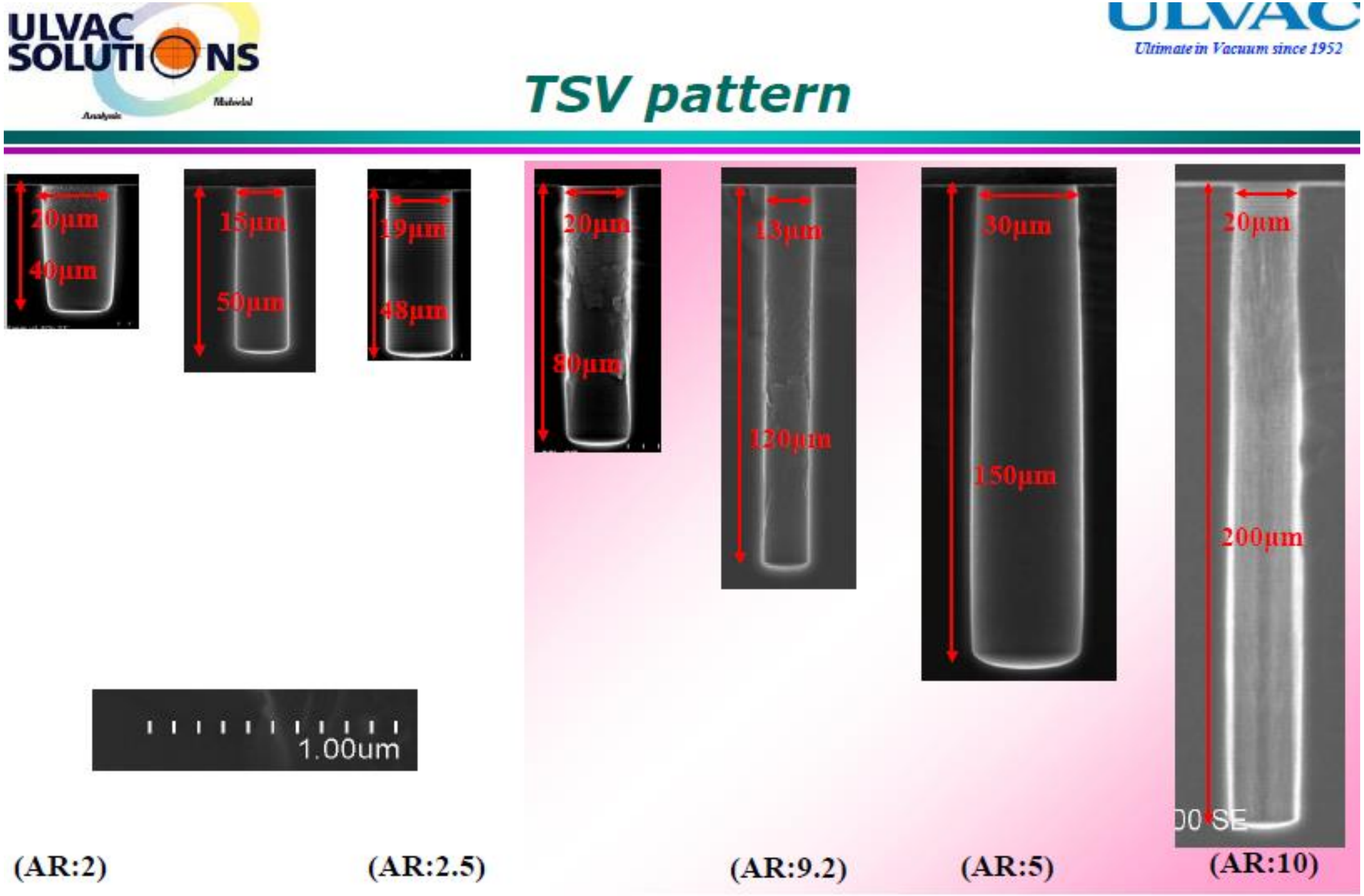
Thin Film Deposition in trench or via

- 准直溅射

Collimated Sputtering



Process Control



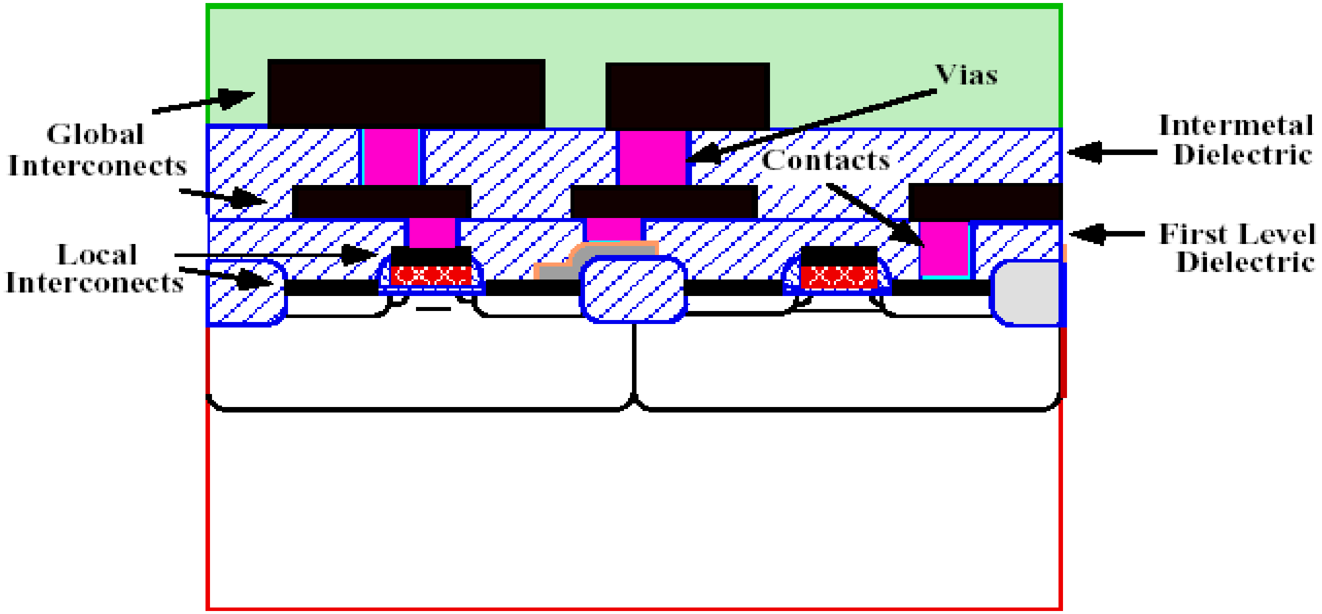
PVD Application

- Salicide
- Barrier
- ARC(anti reflective coating, TiN)
- Interconnect

Application

在分立器件和集成电路制备中用到多种薄膜材料

- 热氧化薄膜 SiO_2
- 电介质膜 SiO_2, Si_3N_4 , BPSG…
- 外延膜 Si, SiGe, GaAs, …
- 多晶硅膜 Poly-Si
- 金属膜Al，Cu，W，Ti，Ni…



Outline

Homework

下图是溅射某薄膜时，沉积速率与靶基距（靶和衬底间的距离）之间的关系。结合溅射原理，解释一下为什么？